

Q8

Given $r_A = 4r_B$, $f_A = f_B = \frac{1}{60}$ Hz,

Using $a_c = r\omega^2$, $\omega = 2\pi f = 0.10467$ rad s⁻¹

$$10 = r_A(2\pi 0.0167)^2$$

$$r_A = 913 \text{ m}$$

$$v_B = r_B\omega = \frac{1}{4}r_A\omega = 23.89 \approx 24 \text{ m s}^{-1}$$

$$a_B = r_B\omega^2 = \frac{1}{4}r_A\omega^2 \approx 2.50 \text{ m s}^{-2}$$

Ans: B

Q9